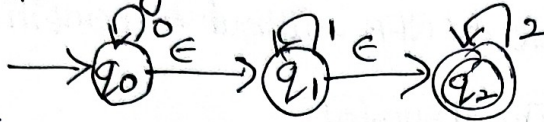


Theory Of Computation

1) Design the following NFA with ϵ -transition to NFA without ϵ -transition.



Solution:-

Step 1:- Find ϵ closure for all state

$$\epsilon\text{-closure of } q_0 = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure of } q_1 = \{q_1, q_2\}$$

$$\epsilon\text{-closure of } q_2 = \{q_2\}$$

Step 2:- Transition table for NFA with ϵ -transition

State	Input symbol			
	0	1	2	ϵ
$\rightarrow q_0$	q_0	ϕ	ϕ	q_1
q_1	ϕ	q_1	ϕ	q_2
q_2	ϕ	ϕ	q_2	ϕ

Step 3:-

Find the transition for the state q_0, q_1, q_2 based on the input symbols 0, 1 & 2.

$$\begin{aligned} \delta(q_0, 0) &= \epsilon\text{-closure}[\delta(\delta^+(q_0, \epsilon), 0)] \\ &= \epsilon\text{-closure}[\delta(q_0, q_1, q_2), 0] \\ &= \epsilon\text{-closure}[\delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0)] \\ &= \epsilon\text{-closure}[q_0] \\ &= \{q_0, q_1, q_2\} \end{aligned}$$

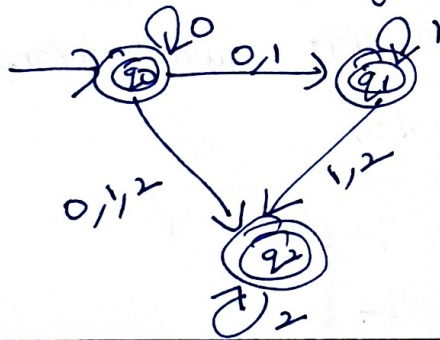
$$\begin{aligned} \delta(q_0, 1) &= \epsilon\text{-closure}[\delta(\delta^+(q_0, \epsilon), 1)] \\ &= \epsilon\text{-closure}[\delta(q_0, q_1, q_2), 1] \\ &= \epsilon\text{-closure}[\delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)] \\ &= \epsilon\text{-closure}[\phi \cup q_1 \cup \phi] \\ &= \{q_1, q_2\} \end{aligned}$$

$$\begin{aligned}
 \delta(q_0, 2) &= \epsilon\text{-closure}[\delta(\delta'(q_0, \epsilon), 2)] \\
 &= \epsilon\text{-closure}[\delta(q_0, q_1, q_2), 2] \\
 &= \epsilon\text{-closure}[\delta(q_0, 2) \cup \delta(q_1, 2) \cup \delta(q_2, 2)] \\
 &= \epsilon\text{-closure}[q_2] \\
 &= \{q_2\}
 \end{aligned}$$

Step 4:- Transition table for NFA without ϵ -transition.

State	Input symbol		
	0	1	2
$\rightarrow q_0$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2\}$
q_1	ϕ	$\{q_1, q_2\}$	$\{q_2\}$
q_2	ϕ	ϕ	$\{q_2\}$

Step 5:- Transition diagram for NFA without ϵ -transition



2) Construct the Minimize DFA for the following regular expression $(a+ab)^*abb$

solution

RE $(a+ab)^*ab$

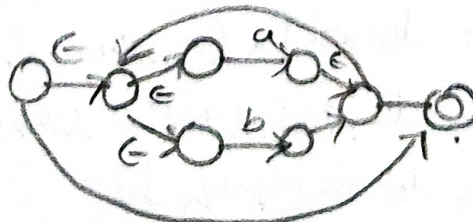
Step 1:-

Construct ENFA diagram from the given RE

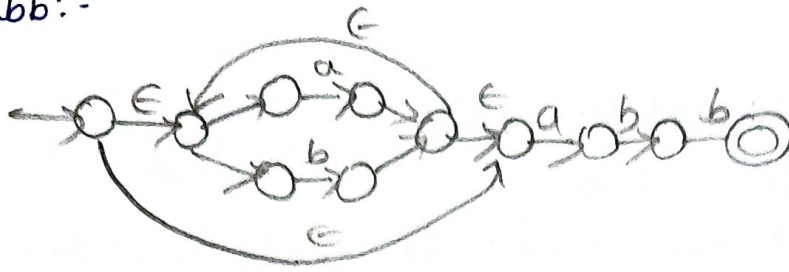
$(a+ab)$:-



$(a+ab)^*$:-



$(a+b)^*abb$:-



Step 2:-

Find the ϵ -closure for the starting state q_0 with inputs a and b

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2, q_3, q_4\} \rightarrow \textcircled{A} \text{ new state}$$

Step 3:-

Find the transition for the new state 'A' based on input symbols a and b

$$\text{move}(A, a) = \{q_3, q_4\}$$

$$\epsilon\text{-closure}[\text{move}(A, a)] = \{q_3, q_4, q_6, q_7, q_1, q_2, q_4\} \rightarrow \textcircled{B} \text{ new state}$$

$$\text{move}(A, b) = \{q_5\}$$

$$\epsilon\text{-closure}[\text{move}(A, b)] = \{q_5, q_6, q_7, q_1, q_2, q_4\} \rightarrow \textcircled{C} \text{ new state}$$

$$\delta(q_0) = \epsilon\text{-closure}[\delta(\delta^1(q_1, \epsilon), 0)]$$

$$= \epsilon\text{-closure}[\delta(q_1, q_2), 0]$$

$$= \epsilon\text{-closure}[\emptyset] \Rightarrow \emptyset$$

$$\delta(q_1, 1) = \epsilon\text{-closure}[\delta(\delta^1(q_1, \epsilon), 1)]$$

$$= \epsilon\text{-closure}[\delta(q_1, q_2), 1]$$

$$= \epsilon\text{-closure}[q_1] \Rightarrow \{q_1, q_2\}$$

$$\delta(q_2, 2) = \epsilon\text{-closure}[\delta(\delta^1(q_1, \epsilon), 2)]$$

$$= \epsilon\text{-closure}[\delta(q_1, q_2), 2]$$

$$= \epsilon\text{-closure}[q_2] \Rightarrow \{q_2\}$$

$$\delta(q_2, 1) = \epsilon\text{-closure}[\delta(\delta^1(q_2, \epsilon), 1)]$$

$$= \epsilon\text{-closure}[\delta(q_2, 1)] \Rightarrow \emptyset$$

$$\delta(q_2, 2) = \epsilon\text{-closure}[\delta(\delta^1(q_2, \epsilon), 2)]$$

$$= \epsilon\text{-closure}[\delta(q_2, 2)] \Rightarrow q_2$$

Step 4:-

Find the transition state for B & C

$$\text{move}(B, a) = \{q_3, q_8\}$$

$$\epsilon\text{-closure}[\text{move}(B, a)] = \{q_1, q_2, q_3, q_4, q_6, q_7, q_8\} \rightarrow B$$

$$\text{move}(B, b) = \{q_5, q_9\}$$

$$\epsilon\text{-closure}[\text{move}(B, b)] = \{q_1, q_2, q_4, q_5, q_6, q_7, q_9\} \rightarrow B$$

$$\text{move}(C, a) = \{q_3, q_8\}$$

$$\epsilon\text{-closure}[\text{move}(C, a)] = \{q_1, q_2, q_3, q_4, q_6, q_7, q_8\} \rightarrow B$$

$$\text{move}(C, b) = \{q_5\}$$

$$\epsilon\text{-closure}[\text{move}(C, b)] = \{q_1, q_2, q_4, q_5, q_6, q_7\} \rightarrow C$$

Step 5:-

To find the 'D'

$$\text{move}(D, a) = \{q_3, q_8\}$$

$$\epsilon\text{-closure}[\text{move}(D, a)] = \{q_1, q_2, q_3, q_4, q_7, q_8, q_6\} \rightarrow B$$

$$\text{move}(D, b) = \{q_5, q_{10}\}$$

$$\epsilon\text{-closure}[\text{move}(D, b)] = \{q_5, q_{10}, q_1, q_2, q_4, q_6, q_7, q_9\}$$

$q_9 \rightarrow E$

Step 6:-

Now E is new state $\text{move}(E, a) = \{q_3, q_8\}$

$$\epsilon\text{-closure}[\text{move}(E, a)] = \{q_1, q_2, q_3, q_4, q_6, q_7, q_8\} \rightarrow B$$

$$\text{move}(E, b) = \{q_5\}$$

$$\epsilon\text{-closure}[\text{move}(E, b)] = \{q_1, q_2, q_4, q_5, q_6, q_7\} \rightarrow E$$

There is no more new states hence transition terminated.

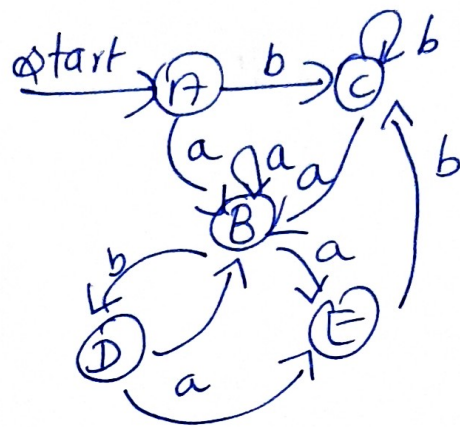
Step 7:-

DFA transition table

State	Input symbols	
	a	b
$\rightarrow A$	B	C
B	B	D
C	B	C
D	B	E
E	B	C

Step 8:-

DFA transition diagram.



DFA minimization:-

$\pi[\text{Final}][\text{Non-Final}]$

Step 1:-

$\pi[E][ABCD]$

Step 2:-

Now group E can't be split further, because it is already a single state

Now consider the group (A,B,C,D) it has four states so need to split it

Now $\delta(A,a)=B$ $\delta(A,b)=C$

$\delta(B,a)=B$ $\delta(B,b)=D$

$\delta(C,a)=E$ $\delta(C,b)=C$

$\delta(D,a)=B$ $\delta(D,b)=E$

$\pi_{\text{new}} = [E][AC][B][D]$

$\pi_{\text{Final}} = \pi_{\text{new}} \Rightarrow [AC][B][D][E]$

Minimized DFA transition table:-

State input symbols
a b

$\rightarrow [AC]$	$[B]$	$[A,C]$
$[B]$	$[B]$	$[D]$
$[D]$	$[B]$	$[E]$
$[E]$	$[B]$	$[A,C]$

